

Coin-Operated Newspaper Vending Machine (FSM Design - VLSI Project)

1. Problem Statement

Design a finite state machine (FSM) for a coin-operated newspaper vending machine. The cost is 15 cents. The machine accepts nickels (5 cents) and dimes (10 cents). Exact or excess payment is allowed, but no change is returned.

2. State Design

States represent accumulated money:

S0 = 0 cents, S5 = 5 cents, S10 = 10 cents

3. State Transition Table

Present State	Input	Next State	Output
S0	N	S5	0
S0	D	S10	0
S5	N	S10	0
S5	D	S0	1
S10	N	S0	1
S10	D	S0	1

4. ASM Chart (Text Form)

```
[S0]
  N -> S5
  D -> S10

[S5]
  N -> S10
  D -> S0 (Z=1)

[S10]
  N -> S0 (Z=1)
  D -> S0 (Z=1)
```

5. Verilog Code

```
module vending_machine (
    input clk,
    input reset,
    input N,
    input D,
    output reg Z
);

reg [1:0] state, next_state;

parameter S0=2'b00, S5=2'b01, S10=2'b10;
```

```

always @(posedge clk or posedge reset) begin
    if(reset) state <= S0;
    else state <= next_state;
end

always @(*) begin
    Z = 0;
    case(state)
        S0: if(N) next_state=S5;
            else if(D) next_state=S10;
            else next_state=S0;

        S5: if(N) next_state=S10;
            else if(D) begin next_state=S0; Z=1; end
            else next_state=S5;

        S10: if(N||D) begin next_state=S0; Z=1; end
            else next_state=S10;
    endcase
end

endmodule

```

6. Testbench

```

module tb;
    reg clk, reset, N, D;
    wire Z;

    vending_machine vm(clk, reset, N, D, Z);

    always #5 clk = ~clk;

    initial begin
        clk=0; reset=1; N=0; D=0;
        #10 reset=0;

        // N+N+N
        #10 N=1; #10 N=0;
        #10 N=1; #10 N=0;
        #10 N=1; #10 N=0;

        // D+N
        #10 D=1; #10 D=0;
        #10 N=1; #10 N=0;

        // D+D
        #10 D=1; #10 D=0;
        #10 D=1; #10 D=0;

        #50 $finish;
    end
endmodule

```

7. Conclusion

The FSM successfully models the vending machine behavior using a Mealy machine. It handles valid coin combinations and dispenses the newspaper correctly.